04-DISPLAYS-AND-INPUT-MODULES

Item #	Item Description	Technical Specifications & Model Details	Required Quantity	Asset Classification
01	**0.96-inch I2C OLED Display Module (SSD1306)**	128×64 monochrome graphic OLED display; SSD1306 driver; 4-pin interface (VCC, GND, SCL, SDA); 3.3V–5V DC; high-contrast text and graphics *(Directly connects to Arduino, ESP32, and Raspberry Pi)*	**15 Units**	**Consumable**
02	**Robocraze 0.91 inch Blue OLED Display Module**	128×32 ultra-compact blue OLED display; I2C/IIC serial interface (VCC, GND, SCL, SDA); 3.3V–5V DC compatible; SSD1306 driver; for Arduino, ESP32, and Raspberry Pi	**10 Units**	**Consumable**
03	**4-Digit 7-Segment Display Module (TM1637)**	0.36-inch 4-digit red display with colon dots; onboard TM1637 driver; 2-wire serial bus (CLK, DIO); 5V/3.3V	**6 Units**	**Consumable**
04	**8×8 LED Dot Matrix Module (MAX7219)**	64 bright red LEDs in 8×8 matrix; MAX7219 serial driver; 5-pin SPI interface; daisy-chainable	**5 Units**	**Consumable**
05	**LDR Light Sensor Module**	5mm photoresistor with onboard LM393 voltage comparator, sensitivity trimmer pot, and dual Digital/Analog outputs (3.3V–5V compatible)	**10 Units**	**Consumable**
06	**Voice Recording & Playback Module (ISD1820)**	ISD1820 chip board; 10-second voice recording; onboard microphone; playback pushbuttons; includes 8Ω 0.5W speaker	**4 Units**	**Consumable**
07	**Dual-Axis Analog Joystick Module**	Dual 10 kΩ potentiometers for X and Y axes with integrated momentary pushbutton switch; spring auto-return to center	**5 Units**	**Consumable**
08	**4× 4 Matrix Membrane Keypad**	16-key matrix keypad (keys 0–9, *, #, A–D); flexible flat ribbon cable with 8-pin 2.54mm pitch female socket	**10 Units**	**Consumable**